



CH₄/O₂ supercritical flame structure and simulation

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ABSTRACT

The oxi-combustion of methane is studied by high-fidelity simulations in the context of liquid rocket engines. The pressure thus exceeds the critical pressure of incoming fluids but stays below the critical pressure of water. Methane and oxygen are injected at cryogenic temperatures in a splitter plate configuration that mimics a small part of a co-axial injector. Simulations are performed with a reduced chemistry containing 17 species and 44 reactions, along with a specific treatment for thermodynamics and transport properties to address real-gas effects. The resulting flame is stuck between two dense flows that locally modify the turbulent kinetic energy. As a consequence, the flame can split and experience different regimes of combustion: non-premixed, lean, and rich premixed. Accordingly, species concentration varies with the flame regime and OH is only found in regions poor in methane, contrary to H₂O. For the latter, depending on the local temperature, a part of it should be in liquid or even solid phase meaning that additional modeling effort should be performed in the future. Large-eddy simulations are conducted with a no-model assumption or using the TFLES approach for the subgrid species source term. The global flame characteristics are recovered but coarsening the mesh makes the rich regime disappear.

1. Introduction

To meet market demands and remain competitive, the next generation of liquid propulsion rocket engines must be reusable, re-ignitable in flight, and capable of operating at different thrust levels [1]. To meet this demand, most space companies have chosen to use the methane/oxygen couple to equip at least the first stage of their launcher, like the Europeans for their future generation of rockets, Ariane Next.

Under rocket engine conditions, CH₄/O₂ combustion occurs at high pressure (supercritical) with cryogenic inlet temperature for liquid oxygen (LOx). The fuel serves as a heat transfer fluid to cool down the walls of the combustion chamber [2] and is often injected hotter than its storage temperature. Reproducing such an extreme environment at the lab scale is therefore particularly challenging but highly valuable for the validation of numerical codes. In the literature, experiments dealing with CH₄/LOx combustion are scarce [3–5] and almost always compared to the reference H₂/LOx flame established with co-axial injectors. For instance, Lux and Haidn [4] have shown that the near-injector CH₄/LOx supercritical flame surrounded the LOx jet on a length equal to 4–5 times the LOx diameter (*d*), from the injection plane, before expanding around 5–6 × *d* because of the high vaporization rate occurring during the LOx jet disintegration, similarly to the H₂/LOx flame. LOx was injected at 120 K and methane at near-ambient temperatures (≈ 270 K). The influence of the injector geometry on

the combustion roughness and stability has been investigated in [5], and an increased flame expansion is observed when using a recessed LOx injector element. The combustion is found smoother but with additional resonant frequencies in the chamber acoustics. In an older study [3], Singla et al. have obtained similar results, i.e. a single flame front attaching the LOx post. However, they also observed a singular flame attachment with a double flame front when oxygen and methane are both injected under transcritical conditions, i.e. when the inlet temperature of each species is less than their respective critical temperature (*T*_{c,O₂} = 154.6 K and *T*_{c,CH₄} = 190.6 K), but for pressure in the combustion chamber (*P*_{ch}) above the critical pressure of each species (*P*_{c,O₂} = 5.05 MPa and *P*_{c,CH₄} = 4.6 MPa). LOx was injected at 85 K and methane at 120 K; *P*_{ch} = 6.0 MPa.

In the above studies, the detection of OH and CH chemiluminescence has been performed to characterize the flame-anchoring zone near the liquid oxygen post-tip. In [6] an order of magnitude is given for the spatial resolution reachable in such experiments. Each pixel of the ICCD camera corresponded to 0.120 mm with a potential degradation due to the finite thickness of the laser sheet. Therefore, the resulting broadening appeared as non-negligible since the structure of the flame edge was estimated around 0.300 mm or less [7,8].

With high-fidelity simulations, the mesh resolution can be less than 1 μm [9] even if the numerical cost becomes prohibitive. Indeed, the

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Table 1

CH₄/LOx supercritical flame simulations. The Turbulent-Chemistry Interaction (TCI) uses either Probability Density Function (PDF) with Tabulated Chemistry (TC) or Fast Chemistry (FC), or kinetic schemes without modeling. Kinetic schemes $X_S \rightarrow Y_R$ have X species and Y elementary reactions plus quasi-steady state species (Q). DNS_ε stands for DNS-like (no subgrid-scale modeling).

Ref.	Config.	Modeling	TCI
[7,20]	Blob	2D DNS _ε	17 _S -44 _R
[21]	Splitter Plate	2D DNS _ε	4 _S -1 _R
[9]	"	2D DNS _ε	TC
[22]	"	2D DNS _ε	9 _S -82 _R -7 _Q
[23]	MASCOTTE	3D LES	PDF + FC
[24]	"	3D LES	PDF + TC
[15]	"	3D LES	PDF + TC
[13]	"	3D LES	9 _S -82 _R -7 _Q

simulation of supercritical flames faces many challenges [10] related to specific phenomena, like real-gas thermodynamics [11] and non-ideal transport fluxes [12], which considerably increase the computational cost compared to simulations dealing with ideal gases. Focusing on CH₄/LOx supercritical flames, numerical simulations, i.e. Direct Numerical Simulation (DNS) or Large-Eddy Simulation (LES), are scarce again in the literature (see Table 1). In [7], a supercritical triple flame is found showing three regimes of combustion, i.e. lean or rich premixed then non-premixed, like in [13,14] where the coupled fluid-wall heat transfer simulation gives a flame that burns in diffusion as well as partially-premixed modes. However, except in [15] in which laminar premixed flames are used to build the tabulated thermochemistry coupled to β -PDF [16,17], other approaches for LES consider a collection of non-premixed flamelets or even no-model, i.e. an implicit diffusion model [14], to describe the CH₄/LOx combustion. Consequently, as the level of resolution of the experiments does not allow a detailed description of the structure of supercritical flames as well as the choice of modeling purely based on diffusion flamelets no longer seems to be so obvious, it becomes necessary to know the exact structure of CH₄/LOx supercritical flames to define the most suitable combustion modeling. To achieve this objective, simulations (DNS-like) are performed in the following on the splitter plate configuration, in a 2D configuration to limit the huge CPU cost inherent to 3D supercritical simulations [18,19].

2. Numerical solver and configuration setup

Simulations are conducted with the numerical code SiTCom-B, already fully detailed in [25,26], which solves the conservation equations of species mass fractions, with complex molecular transport properties, momentum and total sensible energy in their fully compressible form over a structured mesh in a finite volume formulation. The combustion of CH₄ with O₂ is described with the kinetic scheme RAMEC-17S-44R (17 species, 44 reactions) [7,20] which is a reduced version of the detailed mechanism RAMEC (38 species, 190 reactions) [27] dedicated to high-pressure CH₄/O₂ combustion. Such a mechanism can reproduce the structure of triple flames under high-pressure conditions with cryogenic temperatures. The description of real-gas properties required for high-pressure (supercritical) combustion follows the modeling developed in [7,19,20,28,29]: the equation of state is given by the Soave-Redlich-Kwong approach, and transport properties are based on the high-pressure corrections of Chung and Takahashi. When studying supercritical flames, the non-ideal diffusion based on the gradient of the chemical potential [12] can be required to properly describe the flame structure. More precisely, the ordinary diffusion force $\mathbf{d}_k$ of the species k is usually reduced to the gradient of molar fraction as $\mathbf{d}_k = \mathbf{d}_k^{\text{IG}} = \nabla X_k$ instead of being

$$\begin{aligned} \mathbf{d}_k &= \mathbf{d}_k^{\text{IG}} + \mathbf{d}_k^{\text{RG}} \\ &= \nabla X_k + X_k \nabla \mu_k^{\text{sm}} + X_k \frac{W_k h_k}{RT^2} \nabla T, \end{aligned}$$

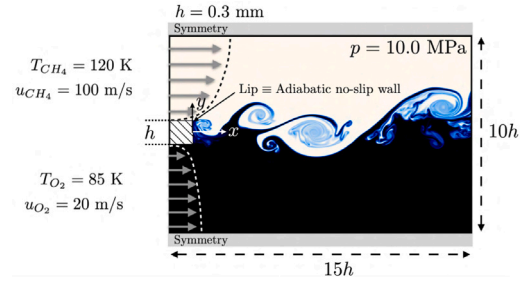


Fig. 1. Description of the 2D numerical benchmark simulated with SiTCom-B. The computational domain is $15h$ in the x -direction and $10h$ in the y -direction. $h = 0.3$ mm.

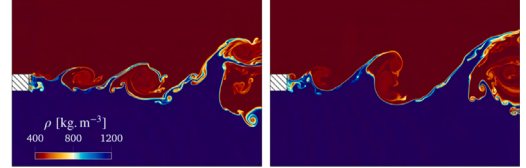


Fig. 2. Numerical simulation (DNS_ε) of the non-reacting splitter plate configuration using ideal (left) and non-ideal (right) diffusion. The density field (ρ) is shown for the case H150.

with $\mathcal{R}$, the perfect gas constant, h_k and W_k , the enthalpy and molar mass of species k , respectively, and T , the temperature. From the computational point of view, the smooth term μ_k^{sm} is a very expensive function depending on the equation of state. Accordingly, considering $\mathbf{d}_k^{\text{RG}}$ in the modeling makes the simulations of 2D or 3D cases unaffordable [19], especially with hydrocarbon flames. Nevertheless, to check the importance of $\mathbf{d}_k^{\text{RG}}$, the mixing of CH₄ with LOx is investigated on the splitter plate configuration with simulation details given in Fig. 1. For the case called H150, the mesh resolution is $\Delta_{\text{H150}} = 2.0 \times 10^{-6}$ m while it is $\Delta_{\text{H30}} = 10.0 \times 10^{-6}$ m for the case H30. For the latter, a LES approach is used as in [28] using the dynamic Smagorinsky model [30,31] and the filtered diffusion force $\tilde{\mathbf{d}}_k$ simply estimated from the filtered fields: $\tilde{\mathbf{d}}_k = \mathbf{d}_k(\tilde{\phi})$ with $\tilde{\phi} \equiv \tilde{T}, \tilde{X}_k, \dots$. Pictures of instantaneous flows are given in Fig. 2. For both cases, classical finger-like structures are recovered with the denser species clearly penetrating in the lighter one. Since it is a 2D simulation, vortex pairing on the methane side appears naturally (Kelvin-Helmholtz mechanism), with the longest finger structures coming from the merging of three vortices. It is also interesting to note that methane has difficulties diffusing inside the LOx stream. In Fig. 3, statistics have been converged on 80 convective times based on the convective velocity (u_c) [32],

$$u_c = \frac{u_{\text{O}_2} + (\rho_{\text{CH}_4}/\rho_{\text{O}_2})^{1/2} u_{\text{CH}_4}}{1 + (\rho_{\text{CH}_4}/\rho_{\text{O}_2})^{1/2}} \approx 50 \text{ m s}^{-1}.$$

If $x/h \leq 7$, for a given mesh resolution H30 or H150, whatever the ideal or non-ideal model for species diffusion used, the mean and standard deviation of density (and other variables) do not show a clear difference. Further downstream at $x/h = 9$, a small impact of the diffusion modeling starts on the LOx side which means that for a longer domain of computation, non-ideal diffusion can have a non-negligible repercussion. Finally, the difference observed in the slope of density in Fig. 3 between H30 and H150 is the consequence of inefficient mixing when the mesh is coarser. As shown in Fig. 4 the penetration of methane inside the oxygen stream is less pronounced in the case of H30 than in H150. Finally, computing H30 with DNS_ε does not change the profile and conclusions.

Based on the above results, $\mathbf{d}_k^{\text{RG}}$ is neglected in the following to save CPU time, knowing that the presence of a flame could change this partial conclusion because the strong temperature (or density) gradient could then have strong consequences on $\mathbf{d}_k$.

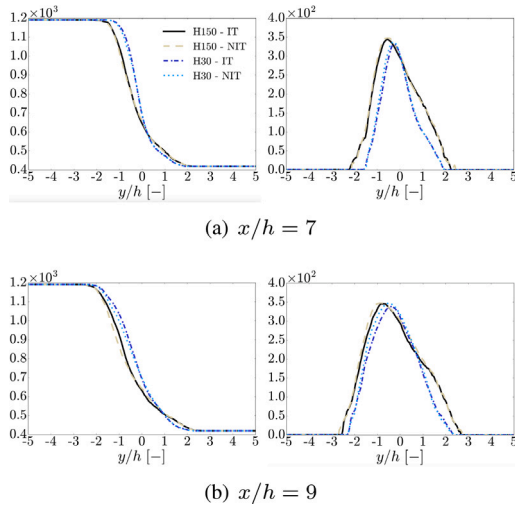


Fig. 3. Mean and standard deviation of density (in kg m^{-3}) for two x/h locations with ideal (IT) and non-ideal (NIT) transport.

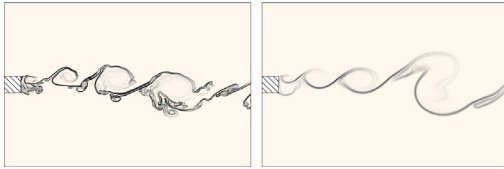


Fig. 4. Instantaneous schlieren fields for CH_4/O_2 mixing. Left: H150 (DNS_r); Right: H30 (LES).

3. High-fidelity flame structure

The configuration of Fig. 2 is fired and run for three convective times to have an installed flame independent from the starting condition before analysis, the flame propagation following ignition having already been studied in [7]. The complexity of the resulting flame is shown in Fig. 5 where the heat release rate is superposed to the mixture fraction (Z). In the present case, the flame develops between two highly dense flows ($\rho_{\text{CH}_4} = 420 \text{ kg m}^{-3}$ and $\rho_{\text{O}_2} = 1200 \text{ kg m}^{-3}$) meaning that the turbulence/flame interaction is restrained to a very short distance not exceeding $3h$. Very thin and very close flame fronts are then observed as in [7], and the overall resulting flame is highly wrinkled. Around $x/h = 6$, pockets of methane penetrate and then burn in the oxygen stream like pockets of oxygen penetrate and then burn in the methane stream. The flame is discontinuous at $x/h = 9$, connecting for a short time the two cold streams of methane and oxygen. As a consequence, triple flame structures can appear locally suggesting the presence of different combustion regimes, premixed and non-premixed. Additional fields are shown in Fig. 6. The flame index [33], defined by

$$\text{F.I.} = \frac{1}{2} \frac{Z - Z_{\text{st}}}{|Z - Z_{\text{st}}|} (1 + \mathcal{T}),$$

with Z_{st} the stoichiometric mixture fraction, and $\mathcal{T}$ the Takeno index [34],

$$\mathcal{T} = \frac{\nabla Y_{\text{CH}_4} \cdot \nabla Y_{\text{O}_2}}{|\nabla Y_{\text{CH}_4} \cdot \nabla Y_{\text{O}_2}|},$$

confirms the presence of different combustion regimes, with F.I. plotted in the flame zone for $\text{HRR} > 0 \text{ W m}^{-3}$. The non-premixed combustion (F.I. = 0) is found behind the splitter plate and close to the stoichiometric isoline which is highly perturbed. The rich premixed combustion regime (F.I. = 1) is mainly observed near the methane stream but with incursions on the oxygen side thanks to vortex pairing.

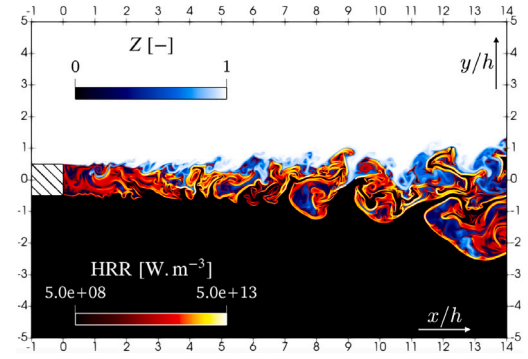


Fig. 5. CH_4/O_2 flame developing with boundary conditions given in Fig. 1. Case H150. The heat release rate (HRR) is superposed to the mixture fraction. $Z = 0$ for oxygen and $Z = 1$ for methane.

Similarly, the lean premixed combustion regime (F.I. = -1) is mainly found close to the oxygen stream. The number of structures detaching from the flame is more important on the methane side than on the oxygen one, as confirmed on the density field, because of the difference in the momentum flux. This feature is illustrated by the momentum ratio (J) [35] between the injected fluids and the resulting flame, characterized by a density of $\rho_F = 5 \text{ kg m}^{-3}$ and the convective velocity, (u_c):

$$J = \frac{\rho_{\text{CH}_4} u_{\text{CH}_4}^2}{\rho_{\text{O}_2} u_{\text{O}_2}^2} = \frac{\rho_{\text{CH}_4} u_{\text{CH}_4}^2}{\rho_F u_c^2} \bigg/ \frac{\rho_{\text{O}_2} u_{\text{O}_2}^2}{\rho_F u_c^2} = \frac{J_{\text{CH}_4}}{J_{\text{O}_2}}$$

In the present case, $J_{\text{CH}_4} \approx 340$ and $J_{\text{O}_2} \approx 38$, which highlights the significant peeling observed near the CH_4 stream than close to the LOx. Finger-like structures can be observed on the fields of density, temperature and oxygen mass fraction. Except behind the splitter plate where the temperature reaches 3800 K because of the adiabatic wall condition, the flame temperature appears not uniform in the burned gases. Note that such a wall condition could cause an overestimation of diffusion fluxes contrary to what is performed in [13]. The overall structure of the turbulent flame is locally composed of various flame fronts distorted and wrinkled by turbulence, and having (Fig. 1) a more intense HRR than in the rest of the flame. Nevertheless, at $x/h = 10$, the flame splits into two parts and the oxygen structure meets the methane stream. Such a behavior comes from the turbulent kinetic energy repartition, which is bounded by the two cold flows that act like a wall condition. In the y -direction, fluid structures coming from the LOx side hit the dense methane streams at high speed, up to $U_y \approx 100 \text{ m s}^{-1}$ i.e. the same order of magnitude as the methane injection velocity (u_{CH_4}), transferring their energy to the $\pm x$ -direction [36]. A relatively large recirculation zone is found behind the splitter plate, on the methane side, which enables the flame attachment to the lip of the injector. $U_x \approx 0 \text{ m s}^{-1}$ near $x/h = 4$. Further downstream, where LOx ligaments deeply penetrate into the reacting zone, the flow accelerates because of the contraction between the two dense cold flows.

In the burned gases, CO_2 is localized throughout the envelope of the turbulent flame. At the same time, CO and H_2 species are found in greater quantities in areas rich in methane (not shown), and OH is found in regions poor in methane, according to the field of flame index. As a consequence, identifying the flame location through the sole concentration of OH could lead to misinterpretations. A solution is to additionally track the evolution of a radical present in methane-rich combustion zones like CH_2O , CH being not present in the reduced kinetic scheme.

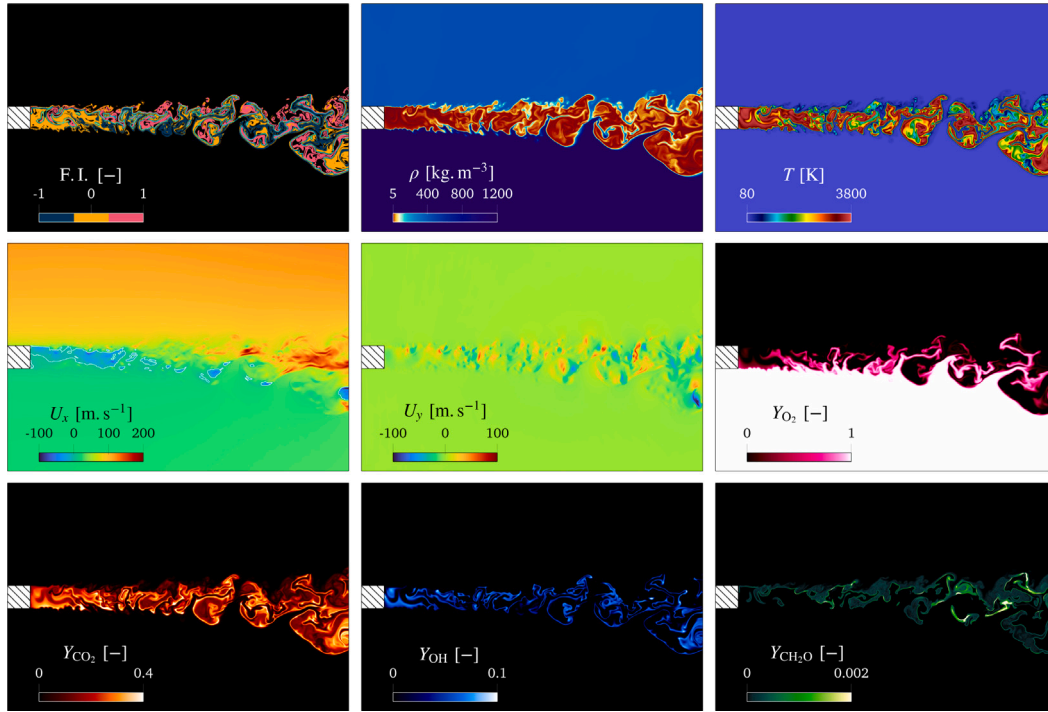


Fig. 6. CH_4/O_2 flame, case H150. F.I. stands for flame index. Other fields uses classical notation for temperature (T), density (ρ), velocity components (U_i), and species mass fractions (Y_k).

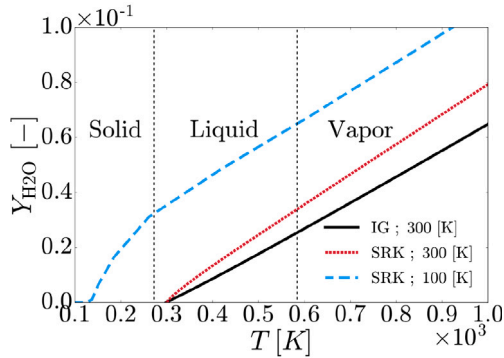


Fig. 7. Mass fraction of H_2O as a function of temperature in a laminar premixed flame at 10 MPa, using the SRK equation of state with an inlet temperature of 300 K or 100 K. $\phi = 1$.

4. Discussion

4.1. Water phases

The case of H_2O is interesting since its critical pressure, $P_c = 22.09$ MPa (with $T_c = 647.3$ K), has a higher value than the pressure of the combustion chamber (10.0 MPa presently). As a consequence, according to the value of temperature, H_2O can be in vapor, liquid, or even solid phase, and would necessitate dedicated modeling as in [37]. To illustrate such behavior, the mass fraction of H_2O coming from the simulation of a 1D laminar stoichiometric premixed flame is plotted in Fig. 7 as a function of temperature. Two inlet temperatures (T_{in}) are considered for CH_4/O_2 mixture, 100 K and 300 K, and the pressure is set to 10 MPa. For $T_{in} = 300$ K, a small but non-negligible amount of H_2O should be found in the liquid state. Switching to $T_{in} = 100$ K increases the quantity of liquid water but also should allow for the creation of solid water. Postprocessing the DNS_c by extracting the mass fraction of H_2O according to the temperatures of phase change at

10 MPa, i.e. $T_{\text{H}_2\text{O}}^{\text{solid}} = 272.4$ K and $T_{\text{H}_2\text{O}}^{\text{liquid}} = 584.15$ K, highlights the areas where H_2O might be in a liquid (Fig. 8(b)) or solid (Fig. 8(c)) states. Unsurprisingly, these phase changes should occur when H_2O meets the two cold flows of methane (120 K) and oxygen (85 K). It corresponds to the envelope of the turbulent flame but seems more conducive to developing on the methane side. Such a result has to be linked with the observations made by Singla et al. [3] on the MASCOTTE bench, in which a block of ice of approximately 4 mm is found attached to the external lip of the coaxial injector, on the methane side, after a test under transcritical injection conditions with temperatures of injection of 85 K for O_2 and 120 K for CH_4 .

4.2. Large-eddy simulation

The flame index that is shown in Fig. 6 highlights how the three regimes of combustion (non-premixed, lean, and rich premixed) are intricated in the flow. Building a subgrid-scale model for turbulent combustion is therefore challenging and considering only diffusion as the preferential modeling may not be sufficient to recover the right flame characteristics. To address this issue, LES are performed on the cases H30 using the dynamic Smagorinsky model [30,31] for turbulence, with, for combustion, either a no-model approach (NO-TCI as in [13,14]) or the dynamically thickened flame LES model (TFLES [38, 39], no efficiency function) conditioned to the premixed combustion regime found in simulations with the Takeno index (T) [34]. The thickening factor is given by $F = 1 + (F_{\text{max}} - 1)S$, with $F_{\text{max}} = 5$. The sensor S uses the production rate ($\dot{\omega}_c$) of the progress variable $Y_c = Y_{\text{CO}_2} + Y_{\text{H}_2\text{O}} + Y_{\text{CO}} + Y_{\text{H}_2}$ in the relation

$$S = \min \left(\frac{|\dot{\omega}_c|}{|\dot{\omega}_c^0|}, 1 \right),$$

where $\dot{\omega}_c^0$ is the maximum of $\dot{\omega}_c$ in the simulation H150. The comparison of the two modeling strategies is provided in Fig. 9. The flame envelope obtained by LES differs from that obtained in the case H150 (Fig. 6), especially on the LOx side, with flames being less turbulent and having a smaller opening angle. In the case H30, a continuous hot

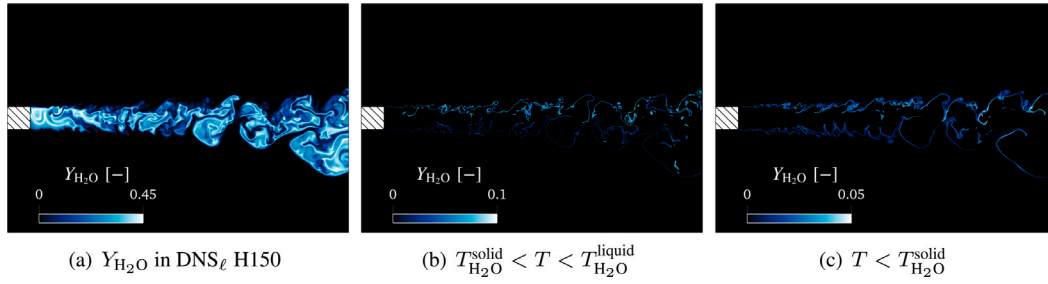


Fig. 8. Mass fraction of H_2O from DNS_ϵ H150 (a) along with postprocessing to highlight the potential liquid (b) or solid (c) water. When $p = 10.0$ MPa, $T_{\text{H}_2\text{O}}^{\text{solid}} = 272.4$ K and $T_{\text{H}_2\text{O}}^{\text{liquid}} = 584.15$ K.

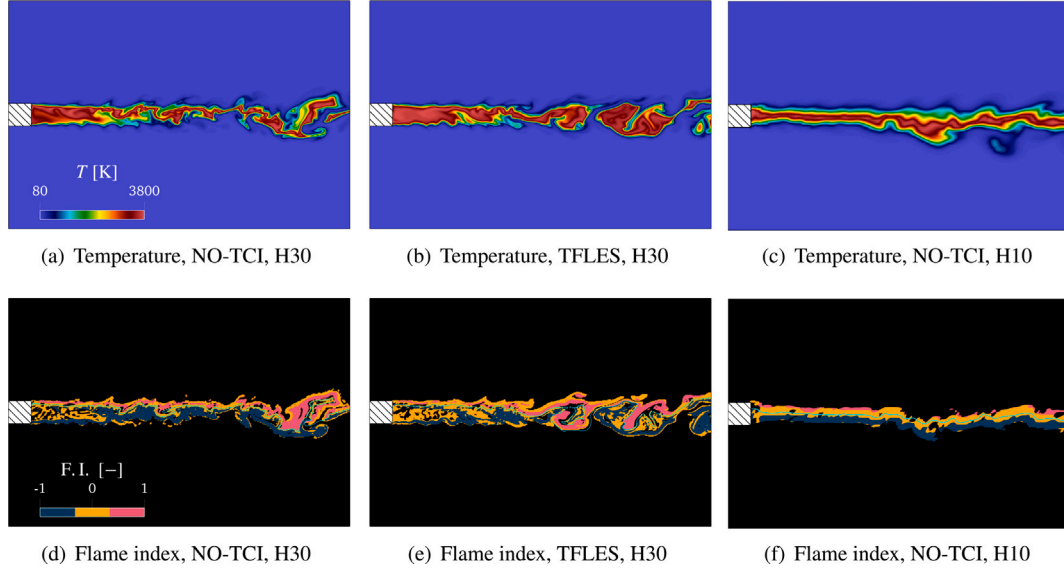


Fig. 9. Instantaneous fields from LES.

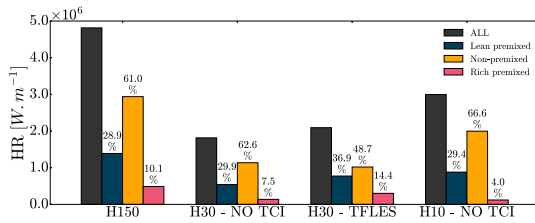


Fig. 10. Distribution of combustion regimes according to the chosen modeling. ALL is the total average heat release for each case.

temperature zone is located behind the LOx post, longer but as thick as in the case H150. However, with a coarser mesh (H10, NO-TCI, $\Delta_{\text{H10}} = 30.0 \times 10^{-6}$ m), such a zone reduces right after the splitter plate, which can strongly modify the subsequent flame behavior. For instance, during the three convective times of the simulation, the discontinuity of the flame front observed in the H150 case was recovered in the H30 case, but not with H10. Overall, the two flames H30 look the same, with the case H30-TFLES slightly hotter and diffused with hot gases more homogeneous than in the H30-NO-TCI case. In addition, the use of NO-TCI yields more oxygen just behind the splitter plate than with TFLES. Both modelings exhibit the three regimes of combustion as in H150, which is not the case anymore with H10 where the rich premixed regime has almost totally disappeared. In the case of H30-TFLES, the thickening factor $\mathcal{F}$ is not activated in the whole flame domain but in locations where premixed flames are the most intense.

In order to quantify the contribution of each combustion regime in the different simulations, the heat release is integrated over time during three convective times. Fig. 10 shows that the average total heat release is greater by more than a factor of two in the H150 case compared to the H30 cases and that the simulation with the TFLES model gives an average total heat release slightly greater than with the NO-TCI model. The no-model assumption promotes the non-premixed regime of combustion whereas TFLES promotes, unsurprisingly, the premixed one.

5. Conclusion

High-fidelity simulations have been performed for the combustion of CH_4/LOx under supercritical conditions, in the case of a splitter plate configuration. The resulting flame is stuck between two dense flows and very intense turbulence–flame interactions are observed. Consequently, pockets of methane can be found in the LOx stream and inversely. The flame therefore burns under various combustion regimes and not as a sole diffusion flame as it is often stated in literature. Coarsening the mesh tends to favor the non-premixed combustion regime which, pushed to the extreme, only reveals a diffusion flame. This study shows that the premixed combustion regime is still present even when simulating a non-premixed splitter plate configuration, which opens new perspectives for modeling the combustion–turbulence interaction. Therefore, combustion modeling under such specific thermodynamic conditions is a real challenge, especially when detailed chemistry is required. Infinitely fast chemistry must therefore be ruled out. The use of TFLES conditioned to the combustion regime can be a promising

strategy if an adapted efficiency function is found to recover the flame length and opening characteristics on a longer computational domain. Finally, the water phases should be included in modeling to state their importance on the flame shape, evolution, and attachment. The origin of the two flame fronts observed in [3] is still an open question, but the variety of combustion regimes along with the potential presence of solid water in the present simulation are certainly major key ingredients. All these results must be consolidated on other operating points through more realistic 3D numerical simulations.

Novelty and significance statement

The novelty of this paper is the highlighting of three combustion regimes (diffusion, rich, and lean premixed) in the case of supercritical CH_4/LOx combustion for rocket engine applications, with the potential creation of liquid or even solid water. This is significant because it changes the way of modeling combustion for LES and therefore the design of the injector/chamber, which is a crucial objective since dealing with such simulations is highly CPU expensive.

CRediT authorship contribution statement

Florian Monnier: Performed research. **Guillaume Ribert:** Wrote the paper. **Louis Duhem-Duvilla:** Performed research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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